# HW 3
# Interest Rates

**Problem 4.5.**
*Suppose that zero interest rates with continuous compounding are as follows:*

| *Maturity (months)* | *Rate (% per annum)* |
|---|---|
| 3 | 8.0 |
| 6 | 8.2 |
| 9 | 8.4 |
| 12 | 8.5 |
| 15 | 8.6 |
| 18 | 8.7 |

*Calculate forward interest rates for the second, third, fourth, fifth, and sixth quarters.*

**Problem 4.8.**
*What does duration tell you about the sensitivity of a bond portfolio to interest rates? What are the limitations of the duration measure?*

**Problem 4.9.**
*What rate of interest with continuous compounding is equivalent to 15% per annum with monthly compounding?*

**Problem 4.11.**
*Suppose that 6-month, 12-month, 18-month, 24-month, and 30-month zero rates are 4%, 4.2%, 4.4%, 4.6%, and 4.8% per annum with continuous compounding respectively. Estimate the cash price of a bond with a face value of 100 that will mature in 30 months and pays a coupon of 4% per annum semiannually.*

**Problem 4.12.**
*A three-year bond provides a coupon of 8% semiannually and has a cash price of 104. What is the bond's yield?*

**Problem 4.14.**
*Suppose that zero interest rates with continuous compounding are as follows:*

| *Maturity( years)* | *Rate (% per annum)* |
|---|---|
| *1* | *2.0* |
| *2* | *3.0* |
| *3* | *3.7* |
| *4* | *4.2* |
| *5* | *4.5* |

*Calculate forward interest rates for the second, third, fourth, and fifth years.*

**Problem 4.16.**
*A 10-year, 8% coupon bond currently sells for $90. A 10-year, 4% coupon bond currently sells for $80. What is the 10-year zero rate? (Hint: Consider taking a long position in two of the 4% coupon bonds and a short position in one of the 8% coupon bonds.)*

**Problem 4.22.**
*A five-year bond with a yield of 11% (continuously compounded) pays an 8% coupon at the end of each year.*

a) *What is the bond's price?*
b) *What is the bond's duration?*
c) *Use the duration to calculate the effect on the bond's price of a 0.2% decrease in its yield.*
d) *Recalculate the bond's price on the basis of a 10.8% per annum yield and verify that the result is in agreement with your answer to (c).*

**Problem 4.34.**
*The following table gives the prices of bonds*

| *Bond Principal ($)* | *Time to Maturity (yrs)* | *Annual Coupon ($)** | *Bond Price ($)* |
|---|---|---|---|
| *100* | *0.5* | *0.0* | *98* |
| *100* | *1.0* | *0.0* | *95* |
| *100* | *1.5* | *6.2* | *101* |
| *100* | *2.0* | *8.0* | *104* |

**Half the stated coupon is paid every six months*

*a) Calculate zero rates for maturities of 6 months, 12 months, 18 months, and 24 months.*
*b) What are the forward rates for the periods: 6 months to 12 months, 12 months to 18 months, 18 months to 24 months?*
*c) What are the 6-month, 12-month, 18-month, and 24-month par yields for bonds that provide semiannual coupon payments?*
*d) Estimate the price and yield of a two-year bond providing a semiannual coupon of 7% per annum.*